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REF/AR

SECTION A · THE HARDWARE

A domestic refrigerator, taken apart

An original 3D model of a household frost-free domestic refrigerator, exploded into its parts. Numbers on each drawing match the legend.

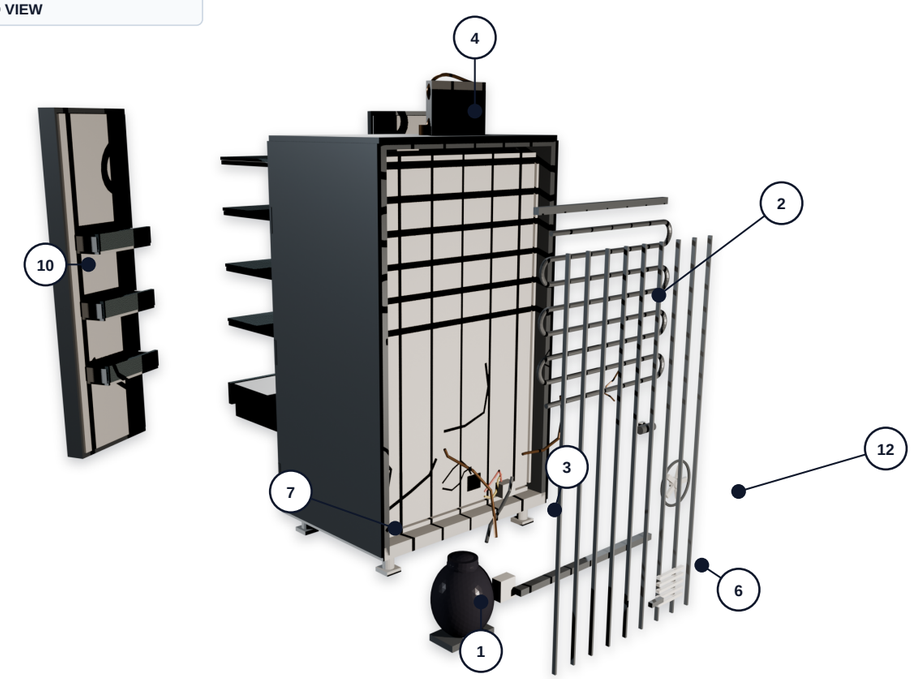
Explore it in 3D
Scan for interactive web explorer & AR view.

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GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

FIG. 1A · ASSEMBLED (OPEN DOORS)



FIG. 1B · EXPLODED VIEW



Component legend

1. **Compressor** — Draws in low-pressure refrigerant vapour from evaporator and raises its ...
2. **Condenser** — Rejects heat (Q_out) from the refrigerant to the surroundings and conden...
3. **Expansion Device** — Throttles high-pressure liquid, plunging its pressure and saturation tem...
4. **Evaporator** — Absorbs heat (Q_in) from the food compartments, vaporising the low-press...
5. **Receiver / Filter-Drier** — Adsorbs trace moisture (zeolite beads) and filters solid particulate deb...
6. **Condenser Fan** — Forces ambient room air across the condenser coil and compressor casing ...
7. **Refrigerant Circuit Lines** — Hot-gas discharge line, liquid line, and insulated low-pressure suction ...
8. **PUF Insulation & Cabinet** — Cyclopentane-blown polyurethane foam insulation (k ≈ 0.02 W/m·K) prevent...
9. **Shelves & Crisper Drawers** — Tempered glass shelves and humidity-regulated vegetable crisper storage ...
10. **Fresh Food Door Assembly** — Insulated cabinet door with magnetic bellows gasket seal and moulded doo...
11. **Freezer Door Assembly** — Dedicated sub-zero compartment door mounted on dual-axis hinge pivots.
12. **Rear Ventilation Grille** — Protective rear stamped grille channelising forced cooling air exhaust a...

Model data

verified

MODEL	Original procedural GLB (glTF 2.0), generated by generate_refrigerator_model.py
VALIDATION	Khronos glTF-Validator: 0 errors, 0 warnings
GEOMETRY	165 named nodes · 151 meshes · 15 PBR materials
FOOTPRINT	1.00 m wide × 0.72 m deep × 1.80 m tall (generic domestic scale)

Typical domestic nameplate specifications

representative

REFRIGERANT	R134a / R600a (Isobutane, GWP=3, ODP=0)
COMPRESSOR	1/6 to 1/4 HP Hermetic Reciprocating (~120W to 180W input)
CONDENSING P	8.0 to 12.5 bar (~35°C to 45°C saturation temp)
EVAPORATING P	1.1 to 1.4 bar (~ -20°C to -15°C saturation temp)
ESTIMATED COP	1.8 to 2.4 (Typical household operating range)

3D model generated procedurally for this project (original geometry, not CAD, not a specific product). Renders from project's three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE HARDWARE: REFRIGERATOR, TAKEN APART	Sahil Panchavishe 612302040 Megh Shidam 612302058	Dr. Shubhangi Deshmukh Dept of Robotics & AI · COEP Tech	1 / 4 SCALE: NTS

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SECTION B · THE REFRIGERATION CYCLE

One loop, two pressures

How refrigerant carries heat out of a cold space: it boils at low pressure in the evaporator and condenses at high pressure in the condenser.

Explore it in 3D
Scan for interactive web explorer & AR view.

FIG. 2A · CYCLE SCHEMATIC

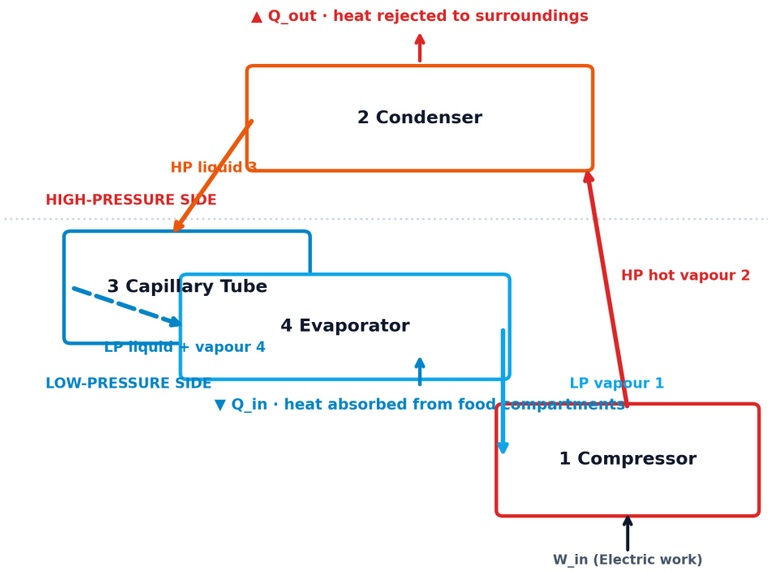
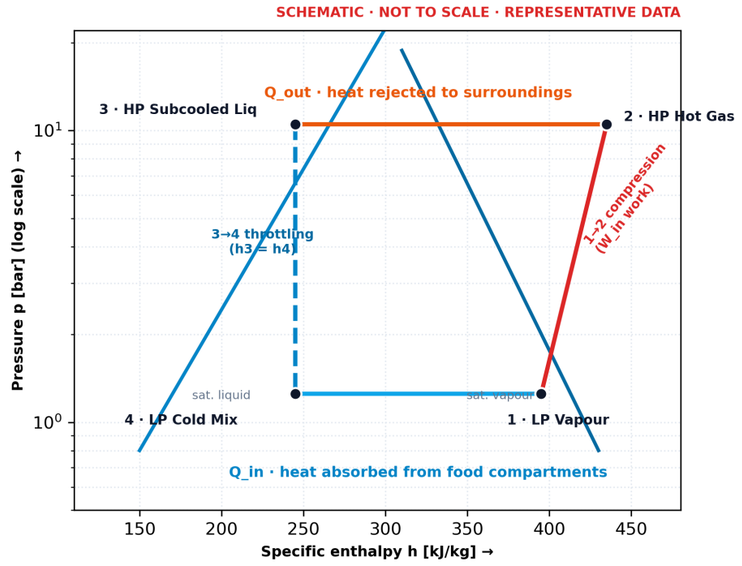


FIG. 2B · p–h SKETCH (SCHEMATIC)



High-pressure side

Compressor outlet → condenser → filter-drier → capillary inlet. Refrigerant rejects heat (Q_out) here.

Low-pressure side

Capillary outlet → evaporator → suction line → compressor inlet. Refrigerant absorbs heat (Q_in).

Energy balance (steady state):

$Q_{out} = Q_{in} + W_{in}$

$COP = Q_{in} / W_{in}$

FIG. 2C · FLOW TRACE IN 3D MODEL



1 Evaporation (heat absorbed)

Cold, low-pressure liquid-vapour mixture enters the evapor...
It absorbs heat from food compartments, boiling into low-p...

Ideal cycle:
Ideal: Constant-pressure heat addition until refrigerant is ...

Real system:
Real: Slight frictional pressure drop across coil; vapour is...

2 Compression (work input)

Compressor draws in low-pressure vapour and squeezes it in...
Driven by electric motor work input. Pressure and saturati...

Ideal cycle:
Ideal: Reversible isentropic (adiabatic) compression (s1 = s...

Real system:
Real: Valve losses, motor winding heat, and piston friction ...

3 Condensation (heat rejected)

Hot, high-pressure vapour discharges into the rear condens...
Rejects heat (Q_out = Q_in + W_in) to room air. Refrigeran...

Ideal cycle:
Ideal: Constant-pressure heat rejection to saturated liquid ...

Real system:
Real: Slight subcooling (3-8 K) occurs at coil exit, ensurin...

4 Expansion (throttling)

High-pressure liquid is forced through the narrow capillar...
High hydraulic resistance causes pressure to drop abruptly...

Ideal cycle:
Ideal: Irreversible throttling at constant enthalpy (h3 = h4...

Real system:
Real: Adiabatic flow with critical choking; soldered to suct...

3D model generated procedurally for this project (original geometry, not CAD, not a specific product). Renders from project's three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE REFRIGERATION CYCLE: THERMODYNAMIC PRINCIPLE	Sahil Panchavishe 612302040 Megh Shidam 612302058	Dr. Shubhangi Deshmukh Dept of Robotics & AI · COEP Tech	2 / 4 SCALE: NTS

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SECTION C · THE PROJECT

One model, three ways in

Objective, architecture and implementation of the Domestic Refrigerator AR Explorer: one procedural model, three ways to learn from it.

Explore it in 3D
Scan for interactive
web explorer & AR view.

C1 Objective & motivation

Students learn the vapour compression cycle as a theoretical four-stage schematic, but a do lation

This project bridges the gap: every component can be tapped to inspect its thermodynamic f place

- Identify the 4 core cycle components in 3D.
- Trace high-pressure and low-pressure sides.
- Walk around the virtual appliance in life-size AR.
- Relate physical layout to the p–h diagram.

C2 Implementation architecture

generate_refrigerator_model.py (Blender)
Python procedural geometry & dual pivots

Domestic_Refrigerator.glb
Khronos Validated glTF 2.0 Asset

Web Explorer
React 19 · Three.js

WebAR Engine

Single shared JSON dataset drives website,
Unity app, and poster for 100% data agreement.

C3 How it works

3D MODEL
Procedural geometry (compressor shell, swept copper tubing, se...
rpentine condenser, evaporator fins) with dual animated door h...

WEB EXPLORER
Interactive tap/click raycast picking, single-part highlight, ...
ghosted isolate mode, explode slider, cycle flow pulse animati...

AUGMENTED REALITY
Zero-install WebAR: WebXR & Scene Viewer on ARCore Android; AR...
Quick Look on iOS. Live camera overlay fallback on desktops &...

UNITY 6 PROJECT
URP scenes: MainMenu, RefrigeratorExplorer, and AR Foundation ...
scene for optional native builds.

C4 Applications & limitations

USE IT FOR
- Pre-lab briefing before thermal lab experiments
- Self-study, revision, and 3D exploration on phones
- Visualizing concealed internal cooling components
- Explaining refrigeration to non-specialists

KEEP IN MIND
- Educational visualization — not manufacturing CAD
- Illustrative generic domestic proportions
- Flow animation is conceptual, not CFD simulation
- No proprietary brand specifications invented
- AR tracking relies on device ARCore / ARKit

FIG. 3A · WEB EXPLORER (DESKTOP)



FIG. 3C · AR PAGE

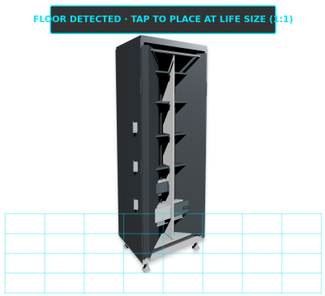
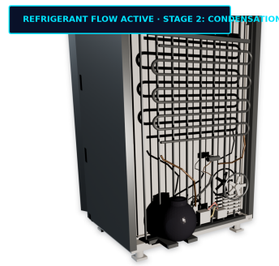


FIG. 3B · PHONE VIEW



FIG. 3D · 3D CYCLE MODE



3D model generated procedurally for this project (original geometry, not CAD, not a specific product). Renders from project's three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE PROJECT: HOW IT IS BUILT	**Sahil Panchavishe 612302040** **Megh Shidam 612302058**	**Dr. Shubhangi Deshmukh** Dept of Robotics & AI · COEP Tech	**3 / 4** SCALE: NTS

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SECTION D · ACCESS

Scan · Explore

Two codes, two ways in. Open your phone camera and point it at either code. Nothing to install.

Explore it in 3D
Scan for interactive
web explorer & AR view.

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GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

AUGMENTED REALITY

CODE 1 · AR

The fridge, in your room

Place the refrigerator on a real floor or table at life size, open the doors, and walk around it.

01 Scan, then tap View in your space.

02 Move the phone slowly over a floor or table.

03 Walk around it; tap Open Doors to see inside.

OR TYPE URL:

https://sahilpanchavishe.github.io/Domestic-Refrigerator-AR-Explorer/#ar

WEB EXPLORER

CODE 2 · WEB

The whole appliance, interactive

Rotate the refrigerator in 3D, tap any of the 13 parts, open doors, and read the specifications.

01 Scan to open the 3D explorer in browser.

02 Tap compressor, condenser, valve or coils.

03 Toggle Open Doors or drag Explode slider.

OR TYPE URL:

https://sahilpanchavishe.github.io/Domestic-Refrigerator-AR-Explorer/

Project

TITLE	Domestic Refrigerator AR Explorer — Interactive 3D & AR
TEAM	**Sahil Panchavishe · 612302040** **Megh Shidam · 612302058**
BRANCH	Robotics & AI (B.Tech)
GUIDE	Dr. Shubhangi Deshmukh
INSTITUTION	Department of Robotics & AI, COEP Technological University, Pune

References & credits

- C. P. Arora, Refrigeration and Air Conditioning, 3rd ed., McGraw-Hill, 2008.
- W. F. Stoecker & J. W. Jones, Refrigeration and Air Conditioning, McGraw-Hill.
- Y. A. Çengel & M. A. Boles, Thermodynamics: An Engineering Approach, McGraw-Hill.
- ASHRAE Handbook — Refrigeration (Systems and Equipment), ASHRAE, 2022.
- Software: three.js, Google <model-viewer>, React 19, Vite 6, TypeScript, ReportLab.
- Unity: Unity 6 (URP, glTFast, AR Foundation, ARCore XR Plugin).
- 3D Model: Original procedural educational visualization created via Blender 5.2 Python API.
- Validation: Khronos Group glTF-Validator 2.0 (0 errors, 0 warnings).

3D model generated procedurally for this project (original geometry, not CAD, not a specific product). Renders from project's three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
SCAN · EXPLORE: ACCESS & CREDITS	**Sahil Panchavishe 612302040** **Megh Shidam 612302058**	**Dr. Shubhangi Deshmukh** Dept of Robotics & AI · COEP Tech	**4 / 4** SCALE: NTS